# Session Objectives

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1. Describe the specific pharmacological treatments for bacterial, viral, and fungal meningitis.
2. Describe the mechanisms of antimicrobials, mainly cell wall inhibitors, aminoglycosides, chloramphenicol, and rifampin. Include the mechanisms for resistance.
3. Describe the mechanism of antivirals which block viral DNA polymerase.
4. Describe the mechanism of antifungals, mainly azoles, flucytosine, and amphotericin.
5. Examine the specific adverse effects of these drugs. Include the management of these effects
6. Describe the role of glucocorticoids as supportive treatment. Include mechanisms and effects.

Source: Course Syllabus

| Microorganism | Standard Therapy | Alternative |
|---|---|---|
| *Streptococcus pneumoniae* | Third generation cephalosporin and vancomycin | Fluoroquinolone |
| *Neisseria meningitides* | Third generation cephalosporin | Chloramphenicol<br>Fluoroquinolone<br>Aztreonam |
| *Listeria* | Ampicillin or penicillin G; with aminoglycoside | SMX/TMP |
| *H. influenzae* | Third generation cephalosporin | Chloramphenicol<br>Cefepime<br>Meropenem<br>Fluoroquinolone |

Group B strep (babies) (*S. agalactiae*); *E coli* (neonates)—Use Beta lactams

*Vaccinate (start at age 11)

# Antimicrobials

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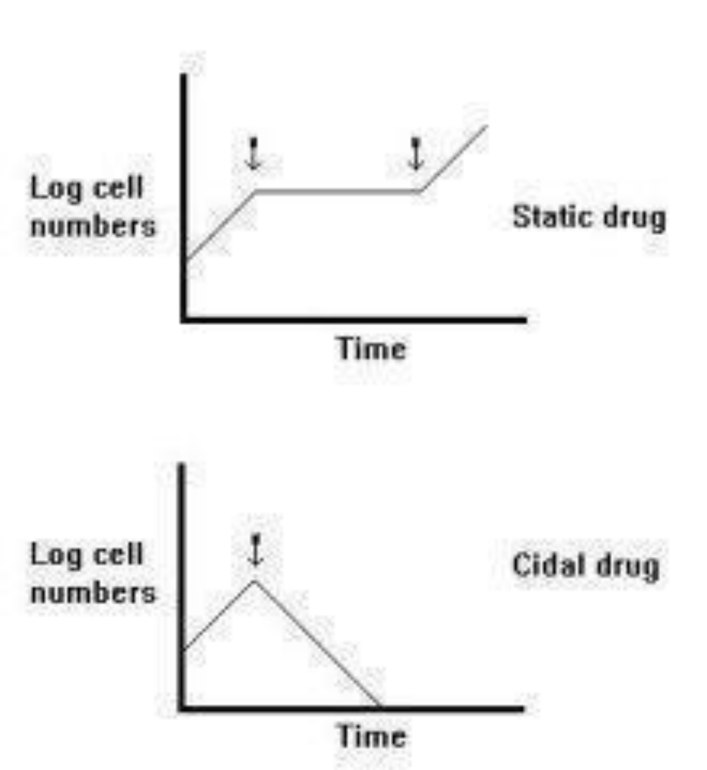
- Bacteriostatic
  - Inhibit the growth of bacteria
  - When used, the duration of therapy must be sufficient to allow cellular and humoral defense mechanisms to eradicate bacteria.
- Bactericidal**
  - Kills bacteria




Source: Course Syllabus

# Classifications

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- Broad Spectrum**
  - Covers both gram positive and gram negative
  - Initial management of meningitis (empiric therapy)*, plus bactericidal.
- Narrow Spectrum
  - Covers only gram positive

**TABLE 20.2 The Spectrum of Activity of Antibiotics and Other Antimicrobial Drugs**

| Prokaryotes | | | | Eukaryotes | | | |
|---|---|---|---|---|---|---|---|
| **Mycobacteria*** | **Gram-Negative Bacteria** | **Gram-Positive Bacteria** | **Chlamydias, Rickettsias†** | **Fungi** | **Protozoa** | **Helminths** | **Viruses** |
| | | ←Penicillin→ | | ←Ketoconazole→ | | ←Niclosamide→ (tapeworms) | |
| ←Streptomycin | Streptomycin→ | | | | ←Mefloquine→ (malaria) | | |
| | | | | | | | ←Acyclovir→ |
| | | | | | | ←Praziquantel→ (flukes) | |
| | ←Tetracycline | Tetracycline | Tetracycline→ | | | | |
| ←Isoniazid→ | | | | | | | |

*Growth of these bacteria frequently occurs within macrophages or tissue structures.
†Obligately intracellular bacteria.

# Cell Wall Inhibitors



- **Bactericidal. Stops Transpeptidation**

  **B- Lactams**

  **Penicillins: most are Gram +, some G-**

  **Cephalosporins: Early drugs are G+, later G+/-**

  **Monobactams (Aztreonam): Gram – (Porin allows bypass of cell surface, therefore no β-lactamase resistance)**

  **Carbapenems (Imipenem and cilastatin): Broad Spectrum**

  **Non B- Lactams**

  **Vancomycin: Gram +**

  **Daptomycin: Gram +**

Source: Course Syllabus

# Resistance

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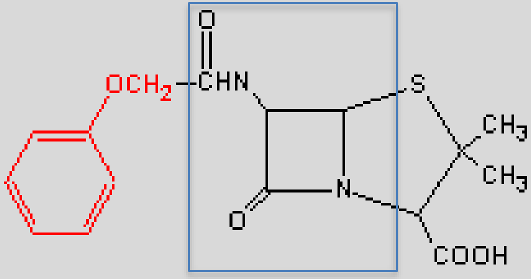
- Degradation by beta lactamases (use “suicide” inhibitors)




- Mutation of penicillin binding proteins (methicillin)
- Porin down-regulation (gram - negative)
- Up-regulation of efflux channels

Source: Course Syllabus

# Penicillin Adverse Effects

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- Anaphylaxis
- Allergy to penicillins (1-2 % cross reactive with cephalosporins)
- Pseudomembranous colitis (ampicillin)
- Probenecid decreases excretion
- Interstitial nephritis
- Drugs which share penicillin hypersensitivity: Penicillins, Cephalosporins, Carbapenems. Similarity in the side chain.
- Drugs which don’t: Aztreonam, Vancomycin

Source: Course Syllabus

# Cephalosporins

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- MOA: as penicillin.
- Inadequate distribution to CNS; except later generations
- Adverse effects: as penicillin; small possibility of cross-reactivity with hypersensitivity
- Drugs with methylthioterazole ring may produce bleeding abnormalities and disulfiram-like reactions.

Source: Course Syllabus

# Cephalosporins

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MOA and adverse effects: As penicillin

1-2% cross reactivity with penicillin

| Generation | Drug | Use |
|---|---|---|
| First | Cefazolin (IV)<br>Cephalexin (po) | mostly gram + infections |
| Second | Cefuroxime (IV, po)<br>Cefoxitin (IV) | *E. coli, Klebsiella, Proteus, H influenzae,* more gram – than the first |
| Third | IV; Ceftriaxone, Ceftazidime | Most gram – and gram +, BBB penetration |
| Fourth | IV, Cefepime | As third generation,more resistant to beta lactamase |

# Vancomycin

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- Inhibits cell wall synthesis by binding 2 d-alanine.
- Bactericidal; gram-positive
- Oral (*C diff*), IV (MRSA, Enterococci)
- Adverse effects: Ototoxicity, Red Man's syndrome (infuse slowly plus use diphenhydramine), Nephrotox.
- Vancomycin-Resistant Enterococci (VRE) happened. D-ala was replaced with D-lactate. What will you use, then???

Source: Course Syllabus

# Cell Wall Synthesis



L-alanine → D-alanine

Alanine Racemase

Cycloserine

vancomycin

Glycopeptide Polymer + 2 D-alanine

Transpeptidase (penicillin, cephalosporin, carbapenem)

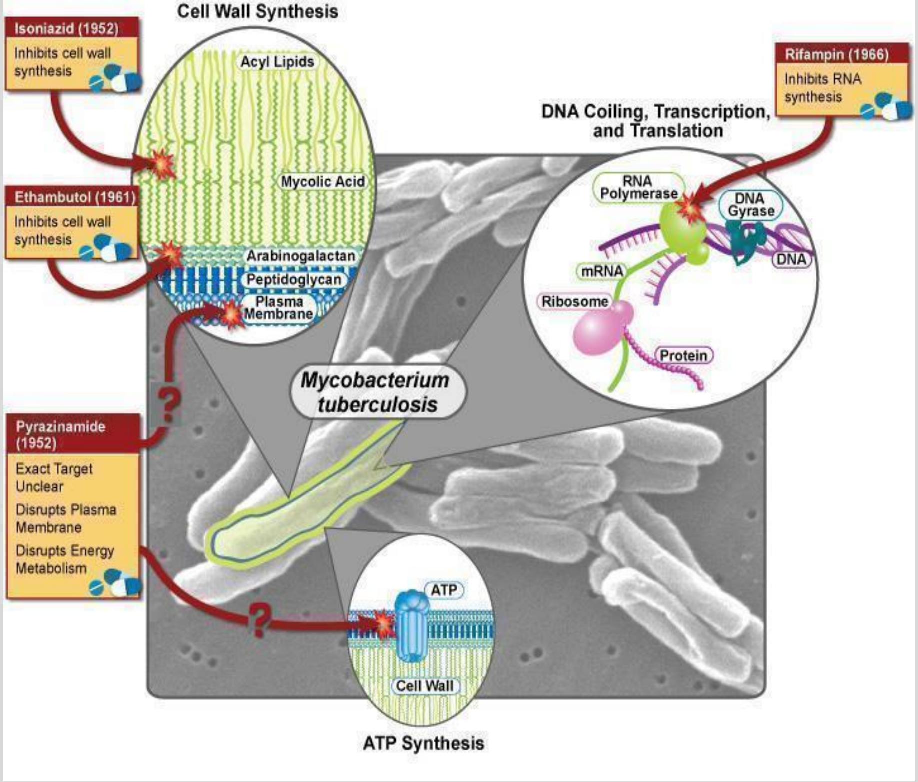
(Penicillin-Binding Proteins)

Cross-linked Proteoglycan Complex

# Rifampin



- Inhibits DNA dependent RNA polymerase
- *Mycobacterium tuberculosis,* meningococcal prophylaxis.
- Adverse effects: Inducer of CYP, hepatotox, and red-orange secretions.
- Resistance: altered structure of enzyme.




Source: Course Syllabus

# Aminoglycoside Adverse Effects

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- <u>Neuromuscular Blockade</u>
- Decreases acetylcholine release from the pre-synaptic neuron
- Avoid in anesthesia with neuromuscular blockade, myasthenia, and Parkinsonism.

- <u>Ototoxicity</u>
- Toxic to both hearing and balance.
- Overlap toxicity with another ototoxic drug (loop diuretics, cisplatin)
- Cranial nerve 8 damage (pregnancy)

Source: Course Syllabus

# Aminoglycoside Adverse Effects



- <u>**Nephrotoxicity**</u>
- **Drug concentrates in proximal tubule.**
- **May cause Acute Tubular Necrosis (ATN). Loss of urine concentrating ability. Non-oliguric renal failure.**
- **Caution in individuals taking other nephrotoxic drugs***
- **Post antibiotic effect: enables once a day dosing.**

Source: Course Syllabus

# Chloramphenicol

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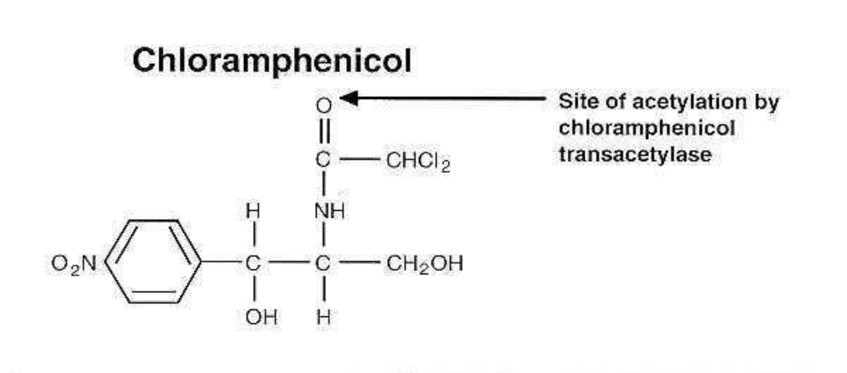
- Broad Spectrum
- Bacteriostatic
- Inhibits peptide formation by binding peptidyltransferase on 50S ribosome




Source: Course Syllabus

- Brain abscess
- Meningitis (*H influenzae, S pneumoniae, N meningitidis*)
- Typhoid
- Rocky Mountain Spotted Fever (*Rickettsial* meningitis doxycycline first line)
- Elimination by glucuronidation
- Wide Distribution: CSF, tears, placenta
- Reduce dose in liver failure.

# Chloramphenicol

- <u>Adverse Effects:</u>
- Bone marrow suppression
- Aplastic anemia
- Gray Baby Syndrome

- <u>Resistance:</u>
- Bacteria are impermeable to drug
- Acetylation of drug

Source: Course Syllabus

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# Tetracycline Adverse Effects

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- GI irritation
- Photosensitivity
- Increased azotemia
- Loss of renal proximal tubule function; Fanconi Syndrome
- Vertigo.  Mainly minocycline.
- *C. difficile* infections
- <u>Resistance</u>: decreased drug uptake into cells or efflux out

# Fluoroquinolones

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- AE: Increased QT interval
- Damage growing cartilage
- Category C
- Tendon rupture
- Photosensitivity
- Drug interactions: CYP 1A2, di, trivalent cations
- Resistance: Reduced uptake of antibiotic, alteration of DNA gyrase

- Ciprofloxacin: broad spectrum
- Levofloxacin, Moxifloxacin
- “Respiratory” *Legionella, Mycoplasma, Chlamydia, Klebsiella, Mycobacterium avium-intracellulare, H. influenzae, S. pneumoniae.*

Source: Course Syllabus

# Viral Meningitis

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- Herpes Simplex I
- St. Louis
- West Nile
- Equine
- Measles, Mumps, Influenza
- Polio

Treatments:

- Acyclovir/Valcyclovir: HSV
- Ganciclovir: CMV
- Neuraminidase Inhibitors: Influenza
- Steroids
- Supportive

Source: Course Syllabus

# Viral Meningitis: Arbovirus

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- 5% of meningitis cases.
- Eastern/Western/Venezuelan equine, St. Louis, California, West Nile, Zika, Dengue.
- Blood-sucking arthropods are the vectors.
- Insect repellant use recommended
- Seizures, Meningocephalitis.
- Treatment is supportive, suppress brain swelling with corticosteroids.

Source: Course Syllabus

# Viral Meningitis

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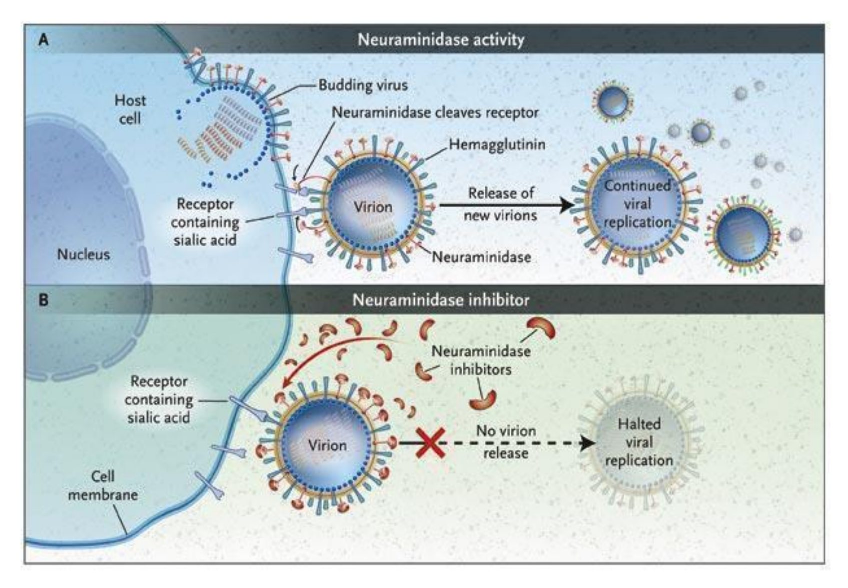
- Mumps causes 20–40% of meningitis cases in undeveloped countries.
- Influenza, mumps, measles:  Vaccinate
- Influenza: Zanamivir, Oseltamivir. Neuraminidase Inhibitors




Source: Course Syllabus

# Herpes



- 4% of viral meningitis
- HSV-2 is most common cause.
- Herpes simplex, Varicella-zoster, Epstein Barr, Cytomegalovirus (CMV)
- Acyclovir is beneficial for most cases
- HAART therapy for HIV patients *
- Ganciclovir reserved for severe CMV infections

Source: Course Syllabus

# Acyclovir

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- IV or topically (Oral HSV, VZV)
- IV use: HSV encephalitis, 1st and recurrent genital HSV, serious HSV or VZV, transplant surgery: HSV reactivation (50%)
- Adverse effects: Nephrotoxicity includes crystallized acyclovir (give hydration), seizures. Valcyclovir (prodrug)
- Foscarnet: phosphorylated; activation not needed (better for resistance). Hypocalcemia, nephrotoxicity.

Source: Course Syllabus

# Ganciclovir

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- Guanosine analog
- Poor bioavailability. Given IV or use valganciclovir
- Uses:
- Cytomegalovirus (CMV) retinitis
- Kaposi's Sarcoma prevention in AIDs.
- 100x more effective than ACV vs CMV
- Adverse effects: Bone marrow suppression
- Resistance: Mutated CMV DNA polymerase or lack of viral kinase. Use foscarnet.

Source: Course Syllabus

# Amphotericin

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- Broad spectrum, fungicidal
- Intrathecal or intraventricular administration (fungal meningitis)
- <u>Resistance:</u> reduced cell membrane ergosterol
- <u>Adverse effects:</u> Infusion “shake and bake” reaction: prophylaxis (cyclooxygenase blockers)
- Renal tubule toxicity (Na, Ca, K, Mg excreted)
- Use liposomal preparations, give other antifungals to lessen toxicity. Hydration

Source: Course Syllabus

# Summary

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- **Describe the specific pharmacological treatments for bacterial, viral, and fungal meningitis.**
- **Describe the mechanisms of these drugs. Include adverse effects.**

Core References:

- **Katzung & Vanderah Basic and Clinical Pharmacology, 16e, 2024; Chapter 43, Beta-Lactam & Other Cell Wall- & Membrane-Active Antibiotics; Chapter 44, Chloramphenicol; Chapter 45, Aminoglycosides; Chapter 48, Antifungals; Chapter 49, Antivirals; Chapter 55, Glucocorticoids.**
- **CNS Infections Summary Guide and Practice Questions (Maria A. Pino, Ph.D)**

Source: Course Syllabus